Effects of porous walls on near-wall supersonic turbulence

Yongkai Chen *
School of Mechanical Engineering, Purdue University, West Lafayette, Indiana 47906, USA

Carlo Scalo
School of Mechanical Engineering and School of Aerospace Engineering, Purdue University, West Lafayette, Indiana 47906, USA

(Received 8 July 2020; accepted 21 July 2021; published 23 August 2021)

We have investigated the effects of the wall permeability on the isothermal-wall supersonic channel flow turbulence. The study is conducted via large-eddy simulations (LES) based on the sub-grid-scale closure proposed by Vreman [Phys. Fluids **16**, 3670 (2004)]. The effects of the wall porosity are modeled via the application of a time-domain impedance boundary condition (TDIBC), which accurately imposes the complex acoustic wall impedance. Bulk Mach numbers of $M_b = 1.50$ and 3.50 are selected, with bulk Reynolds numbers chosen to ensure the same semilocal friction Reynolds number of $\mathrm{Re}_\tau^* \approx 220$. A three-parameter impedance model is used with resonating frequency tuned to the time scales of the energy containing eddies, with wall acoustic resistances $R = 0.50, 1.00, \infty$, ranging from the most permeable to impermeable, respectively. It is found that only cases with $R = 0.50$ yield significant changes in the near-wall turbulence structures, which include a deviation from the linear relation of mean velocity and normalized wall distance in viscous sublayer, and increase in the mean wall-shear stress, as well as a strong increase in both turbulent kinetic energy (TKE) production and dissipation near the wall primarily due to large contribution coming from the instability waves triggered by the permeability. Such waves are found to be confined in the first 10% of the channel half-height near the impedance boundary, creating local circulation zones separated by regions of flow entrainment. It is found that for a given R, the waves are more confined as the Mach number increases. For pressure-related terms, the complex impedance wall effects changes the role of pressure diffusion term in the budget the most, making it responsible for the transport toward the wall, opposite to what is observed in impermeable wall cases. The confined waves also enhance the sink/source effect of the pressure strain term in budget of Reynolds normal stresses, leading to a redistribution of normal stresses.

DOI: 10.1103/PhysRevFluids.6.084607

I. INTRODUCTION

Porous media find widespread applications in aeronautical engineering for noise reduction and passive flow control. Performing high-fidelity numerical simulations including the flow in the pore space requires demanding effort in grid generation and high computational cost associated with the small grid size. When the effect of wall porosity is acoustic in nature, one solution is the mathematical imposition of a complex impedance $\widehat{Z}(\omega)$ at the interface between the flow domain

*chen1305@purdue.edu

 ©2021 American Physical Society

and the porous media denoted as impedance surface. The impedance $\widehat{Z}(\omega)$ takes the form [1]

$$\widehat{Z}(\omega) = \frac{1}{\rho_s a_s} \frac{\widehat{p}(\omega)}{\widehat{\boldsymbol{v}}(\omega) \cdot \boldsymbol{n}} = R(\omega) + iX(\omega), \tag{1}$$

where ω is the angular frequency; $\widehat{p}$ and $\widehat{\boldsymbol{v}}$ are the Fourier transform of the pressure and velocity fluctuations, respectively; $\boldsymbol{n}$ is the unit vector normal to the impedance surface pointing away from the flow domain; ρ_s is the base density and a_s is the base speed of sound, both of which are taken at the impedance surface;i is the imaginary unit $i = \sqrt{-1}$. The impedance itself is a complex number, with the real part $R(\omega)$ and imaginary part $X(\omega)$ called resistance and reactance, respectively. A boundary condition that imposes the complex impedance $\widehat{Z}(\omega)$ is referred to as the impedance boundary condition (IBC). Such IBC formulation assumes linear wave propagation dynamics within the porous media and at the interface with the flow. To implement IBCs in time-domain based flow solvers a time-domain equivalent of Eq. (1), which appears in the form a convolution integral, must be numerically implemented at the impedance surface. Such boundary conditions are called time-domain impedance boundary conditions (TDIBCs). Constraints including passivity, reality, and causality must be honored by the TDIBC implementation to accurately predict the correct physics and to ensure numerical stability, as stated by Rienstra [2] and summarized by Douasbin *et al.* [3]. Tam and Auriault [4] proposed a three-parameter impedance model obtained by fitting experimental data from a typical acoustic liner in the frequency domain. A set of time-domain ODEs were derived directly from it. This type of TDIBC is proved to be well posed provided the imaginary part of the impedance has either nonnegative mass-like reactance or negative spring-like reactance (while using the $e^{-i\omega t}$ convention for Fourier transform). Later, Fung and Ju [5] proposed a TDIBC methodology based on the residue theorem. In this method, the time-domain convolution integral derived from Eq. (1) is evaluated numerically through the trapezoidal rule. However, this formulation is limited to second order accuracy in time. Dragna *et al.* [6] proposed the adoption of auxiliary differential equations (ADE) method for implementing TDIBC, which was originally developed in the field of electromagnetism [7]. With the ADE method, the convolution integral is converted into a set of ODEs with respect to two auxiliary variables, which can be advanced in time with the same time marching technique as the flow solver. Other TDIBCs methods based on different techniques such as z transform [8,9] and characteristic-based formulation [10–12] are also available in the literature. Readers are encouraged to consult the references for more details.

Researchers have applied TDIBCs to help investigate the effect of porous media on a fully subsonic turbulent flow field. For example, Jiménez *et al.* [13] performed Direct Numerical Simulations (DNS) of incompressible turbulent channel flow over one porous wall, in which the boundary conditions are those of purely real impedance, i.e., $v' = -\beta p'$ where β is the porosity coefficient. It was found that permeability significantly alters near-wall turbulence. Local blowing and suction regions have been identified, which on average increase the mean wall-shear stress. Scalo *et al.* [14] conducted a series of numerical experiments of compressible turbulent channel flow over complex wall impedance, using Fung and Ju's TDIBC technique and the three-parameter impedance model with the resonating frequency tuned to the most energetic frequency in the flow. The study spans a flow regime from nearly incompressible ($M_b = 0.05$) to low subsonic ($M_b = 0.5$). Spanwise rollers, exhibiting Kelvin-Helmholtz-type behavior, are found to replace the typical streamwise streaks. These structures are confined in a so-called resonating buffer layer—replacing the classic impermeable buffer layer—and change only the near-wall turbulent structures, as in flow over canopies. Later, [15] performed simulations of a similar flow with the same impedance model, but with the TDIBC formulation proposed by Ref. [4]. The flow conditions were at $M_b = 0.3$, with one case being $M_b = 0.4$, and with various selections of the resonating frequency. Results have shown that the spanwise coherence of the rollers gradually disappear as the resonating frequency is tuned beyond the characteristic flow frequency. All the aforementioned work focus on incompressible or subsonic compressible flow. The current work is motivated by the increasing attention to the application of porous media/coatings in high-speed flow, as well as the desire by the authors to

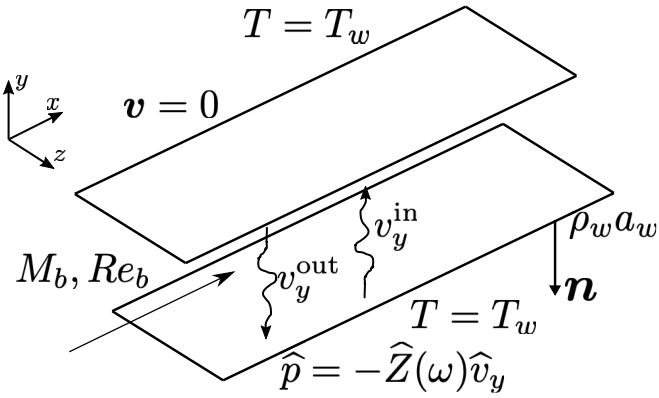

FIG. 1. Sketch of the flow setup for turbulent channel flow over one permeable wall. The negative sign in the impedance boundary condition comes from the convention of the outward-facing normal $\boldsymbol{n}$ [see Eq. (1)].

extend the work by Scalo *et al.* [14] to the supersonic and hypersonic regime. The effect of such surfaces on the high-speed near-wall turbulence is of great interest. To the authors' knowledge, there is no existing published work about their applications in high-speed turbulent flows. This paper and its companion publication [16] pioneer such investigation. The paper is organized as follows: Section II describes the flow problem, Sec. III provides the numerical setup, results are discussed in Secs. IV–VI, and conclusions are provided in Sec. VII.

II. PROBLEM FORMULATION

In this paper, large-eddy simulations (LES) of fully developed turbulent channel flow with isothermal walls are performed. The top wall is kept impermeable for all cases, while the bottom wall is allowed to have various degrees of permeability modeled by time-domain impedance boundary conditions. The flow setup is shown in Fig. 1. For clarity, the coordinate $(x, y, z) = (x_1, x_2, x_3)$ represent the streamwise, wall-normal, and spanwise directions, respectively. The reference parameters are the channel half-height δ, speed of sound at the wall a_w, wall temperature T_w, and bulk density $\rho_b = \langle\rho\rangle_V$, where the brackets $\langle\cdot\rangle_V$ indicate the volumetric-average operator. The bulk Reynolds number is then defined as

$$\mathrm{Re}_b = \frac{\rho_b U_b \delta}{\mu_{\text{ref}}}, \tag{2}$$

where $U_b = \langle\rho u_1\rangle_V/\rho_b$ is the density weighted bulk velocity, also yielding the bulk Mach number $M_b = U_b/a_w$; μ_{ref} is the dynamic viscosity taken at the wall. In the current work, the bulk Mach number M_b is chosen as $M_b = \{1.50, 3.50\}$. Different bulk Reynolds numbers Re_b are selected so that all Mach number cases share a similar viscous Reynolds number $\mathrm{Re}_\tau^* \approx 220$, that is, the semilocal viscous Reynolds number accounting for variable density effects [17] present in this flow.

The simulations are performed by solving the compressible Navier-Stokes equations:

$$\frac{\partial\rho}{\partial t} + \frac{\partial\rho u_j}{\partial x_j} = 0, \tag{3}$$

$$\frac{\partial\rho u_i}{\partial t} + \frac{\partial\rho u_i u_j}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial\tau_{ij}}{\partial x_j} + \rho f_i, \tag{4}$$

$$\frac{\partial\rho E}{\partial t} + \frac{\partial\rho u_j E}{\partial x_j} = -\frac{\partial p u_j}{\partial x_j} + \frac{\partial(u_k\tau_{kj} - q_j)}{\partial x_j} + \rho f_j u_j. \tag{5}$$

Here τ_{ij} is the stress tensor given by

$$\tau_{ij} = (\mu + \mu_{\text{sgs}})\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial u_k}{\partial x_k}\right), \tag{6}$$

with the subscript "sgs" denoting sub-grid-scale (SGS) model terms. Note that a repeated i index implies the summation when used as a subscript, while used in equations such as Eq. (1) it indicates the imaginary unit. The heat flux vector q_j is defined as

$$q_j = -C_p\left(\frac{\mu}{\text{Pr}} + \frac{\mu_{\text{sgs}}}{\text{Pr}_{\text{sgs}}}\right)\frac{\partial T}{\partial x_j}, \tag{7}$$

where C_p is the heat capacity at constant pressure and $\text{Pr} = 0.72$ is the Prandtl number. The turbulent Prandtl number Pr_{sgs} is chosen to be 0.90. The ideal gas constitutive relation $p = \rho R_{\text{gas}} T$ is used with R_{gas} being the gas constant. The power law $\mu/\mu_{\text{ref}} = (T/T_{\text{ref}})^n$ is assumed for dynamic viscosity with $n = 0.76$. The eddy viscosity model by Vreman [18] is used as the turbulence closure.

To drive the flow, a uniform external body force f_1 in streamwise direction is dynamically adjusted to maintain the desired bulk mass flow rate. Its time average reads

$$\overline{f}_1 = \frac{\tau_w}{\delta \rho_b}, \tag{8}$$

where τ_w is the mean wall shear stress. For all simulations, periodic boundary conditions are used in streamwise and spanwise directions. For wall boundaries, the conditions are given as

$$u_1 = u_3 = 0, \tag{9a}$$

$$\begin{cases} u_2 = 0, & \text{impermeable}, \\ \hat{p} = \rho_w a_w \widehat{Z}(\omega)\hat{\boldsymbol{v}} \cdot \boldsymbol{n}, & \text{permeable}, \end{cases} \tag{9b}$$

$$T = T_w, \tag{9c}$$

where $T_w = 1$ and $a_w = 1$ based on current normalization. For both permeable and impermeable walls, no-slip conditions are used for the tangential velocity components. One underlying assumption is that the orifice size of a corresponding physical array of Helmholtz resonators in the current setup is very small compared to the channel half height δ, and thus yields negligible tangential velocities, as also assumed for other types of porous walls used in high-speed flow applications [19]. Appendix A provides an analysis of the range of pore sizes relative to the boundary layer thickness typically found in high-speed flow conditions, supporting such assumption. For purely impermeable walls, the no-penetration condition is adopted (i.e., infinite impedance), while for permeable walls, TDIBC is applied with prescribed finite wall impedance $\widehat{Z}(\omega)$. Note that for the permeable wall simulations, the net mass flux across the impedance boundary is not enforced to be numerically zero. To prevent the occurrence of spurious net mass accumulation or leak in the channel, the surface-average mass flux through the impedance boundary is calculated at each sub-timestep, and converted into a density weighted wall-normal velocity $\tilde{v} = \overline{\rho v}/\overline{\rho}$. Such velocity is then subtracted from the instantaneous vertical velocity at the impedance boundary, ensuring a zero net mass flux through the whole impedance boundary at each sub-timestep. More details on the TDIBC implementation are discussed in the next two subsections.

A. TDIBC based on ADE method

Instead of directly using $\widehat{Z}(\omega)$, the TDIBC in the current work relies on the time-domain expression of the complex wall softness coefficient $\widehat{S}(\omega)$ defined as

$$\widehat{S}(\omega) = \frac{2}{1 + \widehat{Z}(\omega)}. \tag{10}$$

Its relation with the complex wall reflection coefficient, $\widehat{R}(\omega)$ is simply $\widehat{S}(\omega) = \widehat{R}(\omega) + 1$. The reflection coefficient $\widehat{R}(\omega)$ directly measures the magnitude and phase changes between the incident and the reflected waves, with values of $-1, 0, +1$ corresponding to surfaces of hard reflection, no reflection and pressure-release conditions, respectively. The complex wall-softness shares similar properties and relates the (flow) domain-leaving wave, $\hat{v}_n^{\text{out}}(\omega)$, to the domain-entering wave, $\hat{v}_n^{\text{in}}(\omega)$, both normal to the bottom-wall impedance surface, via

$$\hat{v}_n^{\text{in}}(\omega) = [\widehat{S}(\omega) - 1)]\hat{v}_n^{\text{out}}(\omega). \tag{11}$$

The time-domain expressions for such waves at the bottom boundary (with $\boldsymbol{n}$ for the bottom boundary pointing downwards) read

$$v_n^{\text{out}}(t) = v_n'(t) - \frac{p_w'(t)}{\rho_w a_w}, \tag{12a}$$

$$v_n^{\text{in}}(t) = v_n'(t) + \frac{p_w'(t)}{\rho_w a_w}, \tag{12b}$$

where $p_w'(t)$ is the pressure fluctuation at the wall. Such local recasting of the fluctuating field is consistent with Poinsot and Lele [20]'s characteristic boundary condition approach for the compressible Navier-Stokes equations. The impedance is here after assumed to act on only the acoustic waves.

The time-domain equivalent form of Eq. (11) is

$$v_n^{\text{in}}(t) = -v_n^{\text{out}}(t) + \int_{-\infty}^{\infty} s(\tau) v_n^{\text{out}}(t-\tau) d\tau. \tag{13}$$

The goal of TDIBC methods is to evaluate this integral while ensuring the physical realizability of the boundary condition. It has been summarized by Rienstra [2] that physical realizability can be ensured by satisfying passivity, reality, and causality constraints. However, such constraints are derived based on $\widehat{Z}(\omega)$, i.e., the formulation in Eq. (1). When the characteristic form of the incoming and outgoing acoustic waves is used [Eq. (11)], the constraints, which should be applied on $\widehat{S}(\omega)$ to ensure a causal integral, read:

(1) *Passivity of the impedance boundary.* The acoustic power through the impedance surface must entail a net acoustic energy transfer outside of the domain. Suppose $\widehat{S}(\omega)$ can be written as a fractional function, i.e., $\widehat{S}(\omega) = \widehat{A}(\omega)/\widehat{B}(\omega)$, then such constraint requires

$$\Re\{\widehat{B}(\omega)\} \geqslant 1, \forall \omega \in \mathbb{R}, \tag{14}$$

which is equivalent to $\Re\{\widehat{Z}(\omega)\} \geqslant 0$ for all real ω.

(2) *Reality of the signals.* This guarantees the signals obtained in time-domain are purely real, resulting in

$$\widehat{S}(\omega) = \widehat{S}^{\star}(-\omega), \tag{15}$$

with $\star$ representing the complex conjugate.

(3) *Causality.* The state of the physical process at the current time should not depend on the information from future. This means that the contribution to the convolution integral in Eq. (13) due to $\tau < 0$ should be zero, hence $s(\tau) = 0$ for $\tau < 0$.

The TDIBC methodology adopted in the current work builds upon the auxiliary differential equation (ADE) method [6,7] as well as the recasting of $\widehat{S}(\omega)$ into the summation of partial fractions using its residues, $\mu_k = a_k + ib_k$, and poles, $p_k = c_k + id_k$, which is interpretable as the linear superposition of n_0 causal harmonic oscillators [21],

$$\widehat{S}(\omega) = \sum_{k=1}^{n_0} \left(\frac{\mu_k}{i\omega - p_k} + \frac{\mu_k^{\star}}{i\omega - p_k^{\star}} \right). \tag{16}$$

With such form, the reality constraint is automatically satisfied. The causality constraint requires the real part of the poles p_k to be negative, i.e., $c_k \leqslant 0$, which then yields the expression in the time domain,

$$s(t) = \sum_{k=1}^{n_0} \left(\mu_k e^{p_k t} + \mu_k^\star e^{p_k^\star t}\right) H(t), \tag{17}$$

where $H(t)$ is the Heaviside function resulting from the inverse transform. Note that this mathematical form only implies one-way causality, consistent with the fact that the acoustic characteristic reflected off the impedance boundary $v_n^{\text{in}}(t)$ is dependent, in a one-way fashion, on $v_n^{\text{out}}(t)$, impinging on the impedance boundary.

A new set of auxiliary variables is then defined,

$$\psi_k^{(1)}(t) = \int_0^t 2a_k e^{c_k(t-\tau)} \cos[d_k(t-\tau)] v_n^{\text{out}}(\tau) d\tau, \tag{18}$$

$$\psi_k^{(2)}(t) = \int_0^t 2b_k e^{c_k(t-\tau)} \sin[d_k(t-\tau)] v_n^{\text{out}}(\tau) d\tau, \tag{19}$$

and the domain-leaving, $v_n^{\text{out}}(t)$, and domain-entering, $v_n^{\text{in}}(t)$, waves can be related via

$$v_n^{\text{in}}(t) = -v_n^{\text{out}}(t) + \sum_{k=1}^{n_0} 2\left[a_k \psi_k^{(1)}(t) - b_k \psi_k^{(2)}\right], \tag{20}$$

with $\psi_k^{(1)}(t)$ and $\psi_k^{(2)}(t)$ governed by the ODEs

$$\frac{d\psi_k^{(1)}(t)}{dt} = c_{k-1}\psi_k^{(1)}(t) - d_{k-1}\psi_k^{(2)}(t) + v_n^{\text{out}}(t), \tag{21}$$

$$\frac{d\psi_k^{(2)}(t)}{dt} = c_{k-1}\psi_k^{(2)}(t) + d_{k-1}\psi_k^{(1)}(t), \tag{22}$$

where a_k, b_k, c_k, d_k are the real and imaginary parts of the residues μ_k and poles p_k, as stated above. Such ODEs can then be advanced with the same temporal scheme as the main flow solver and boundary conditions can be applied by solving for v_n and p'_w from Eq. (12). A more generalized impedance boundary condition could be developed to account for vortical and entropy mode impedances (see Suzuki and Lele [22]). This effort is out of scope of the current paper and it would also raise issues of practical realizability of such generalized boundary condition.

B. The three-parameter impedance model

In the present work, the three-parameter impedance model proposed by Tam and Auriault [4] is adopted:

$$\widehat{Z}(\omega) = R + i(X_+\omega - X_{-1}\omega^{-1}), \tag{23}$$

which can be shown to be equivalent to a single-pole Helmholtz oscillator [21]. Here R, X_+, X_{-1} are acoustic resistance, acoustic mass-like reactance, and acoustic spring-like reactance, respectively. Note that unless otherwise indicated, the time harmonic convention $e^{+i\omega t}$ is assumed in this paper, as opposed to Tam and Auriault [4] who adopted the $e^{-i\omega t}$ convention. The model in Eq. (23) entails a frequency-selective acoustic permeability, which can be tuned to a characteristic frequency ω_{res} given by

$$\omega_{\text{res}} = \sqrt{\frac{X_{-1}}{X_{+1}}}. \tag{24}$$

In this paper, an equivalent set of three parameters R, ω_{res} and ζ are used where ζ is the damping ratio defined as

$$\zeta = \frac{1+R}{2\omega_{\text{res}}X_{+1}}. \tag{25}$$

The impedance model in Eq. (23) can then be recast into the fully dimensionless form

$$\widehat{Z}\left(\frac{\omega}{\omega_{\text{res}}}\right) = R + i\frac{R+1}{2\,\zeta}\left[\frac{\omega}{\omega_{\text{res}}} - \frac{\omega_{\text{res}}}{\omega}\right]. \tag{26}$$

With the above form of impedance and its connection to $\widehat{S}(\omega)$, one can show that the three aforementioned constraints in Sec. II A requires (for an underdamped system)

$$\begin{cases} R \geqslant 0, & \text{passivity,} \\ \zeta \in (0,1), & \text{causality,} \end{cases} \tag{27}$$

where the reality constraint Eq. (15) is automatically satisfied. Preliminary results have shown that a very low resistance R—or, equivalently, very high permeability—could cause high amplitude instability waves in the flow and thus impose severe constraints on the allowable Courant-Friedrichs-Lewy (CFL) numbers. Hence, values of R are selected as $R = 0.50, 1.00, \infty$ for both Mach numbers, where the infinite value of R corresponds to impermeable wall conditions. A value of $\zeta = 0.5$ is selected for all permeable wall simulations. The impedance boundary in the current work is designed to react to the frequencies of energy containing eddies of the overlying turbulent flow, which yields the tuning condition

$$\omega_{\text{res}} = 2\pi U_b/\delta. \tag{28}$$

consistent with the approach followed by Scalo *et al.* [14]. It should be noted that the TDIBC based on such three-parameter impedance model could be implemented in a simpler and more straightforward way, such as the method used by Ref. [23]. In the current work, however, we have chosen to adopt a general approach that can allow for any realizable form of impedance to be applied as a boundary condition in a high-fidelity compressible-flow calculation.

III. NUMERICAL SETUP AND FLOW CONDITIONS

The compressible Navier-Stokes equations with prescribed boundary conditions are solved via a six-order compact finite differencing code originally developed by Ref. [24] and now under continued development at Purdue. The code implements grid transformation to allow the problem to be solved on curvilinear grids. A semi-implicit temporal scheme is used to alleviate the constraints in CFL numbers imposed by strong vertical velocities near the permeable wall, and it has been developed based on previously established fully implicit methods [24–27] and the explicit Runge-Kutta (RK) scheme used by Ref. [28]. Only the wall normal derivative terms are treated implicitly and are solved within each Runge-Kutta substep. The approximate factorization allows the resulting matrix-inversion problem to be solved iteratively with relatively low cost by using alternating direction implicit (ADI) sweeping. Three subiterations have been proven to be sufficient to ensure convergence at each RK substep. The validation of such scheme can be found in the reference [16].

The flow conditions and the grid size of all cases presented are listed in Table I. The domain size for cases with impermeable walls are both $L_x \times L_y \times L_z = 12\delta \times 2\delta \times 4\delta$, while the rest of the cases with finite R can be appropriately simulated in a shorter domain in the streamwise direction, $L_x \times L_y \times L_z = 8\delta \times 2\delta \times 4\delta$, as proven by the two-points correlations given in Appendix B. The number of grid points is chosen so that the grid spacing is fine enough to resolve the near-wall event induced by the impedance wall. A grid sensitivity study has been performed for the case with most energetic response, i.e., $M_b = 3.50$, $R = 0.50$, and the results are given in Appendix C. The validation of TDIBC implementation can be found in the companion publication [16]. In Table II

TABLE I. Flow conditions, impedance parameters, and number of grid points for all the LES cases. Cases run at $M_b = 1.50$ and 3.50 are hereafter labeled with the "(I)" and "(II)" Roman numerals, respectively.

Group	M_b	Re_b	R	$L_x \times L_y \times L_z$	$N_x \times N_y \times N_z$
(I)	1.50	5000	∞	$12\delta \times 2\delta \times 4\delta$	$160 \times 192 \times 120$
			1.00	$8\delta \times 2\delta \times 4\delta$	$128 \times 192 \times 128$
			0.50	$8\delta \times 2\delta \times 4\delta$	$128 \times 192 \times 128$
(II)	3.50	10000	∞	$12\delta \times 2\delta \times 4\delta$	$400 \times 256 \times 240$
			1.00	$8\delta \times 2\delta \times 4\delta$	$270 \times 256 \times 240$
			0.50	$8\delta \times 2\delta \times 4\delta$	$340 \times 300 \times 300$

the grid resolution is reported in terms of the plus units, along with the time-step and CPU hours per flow-through time. Note that due to the asymmetry in the boundary conditions, data based on top and bottom wall statistics are given separately.

IV. MEAN VELOCITY AND REYNOLDS STRESSES

In this section, some of the first and second-order mean flow statistics are reported. Unless otherwise stated, the Favre averaging [30] is used. The Favre-averaged mean value of a quantity ϕ is denoted by $\widetilde{\phi}$ and the fluctuation by ϕ'', while for Reynolds average $\overline{\phi}$ and ϕ' are used to represent the mean and the fluctuation part, respectively. Figure 2 shows the transformed mean streamwise velocity profile as well as the Favre-averaged Reynolds stresses $\widetilde{u_i''u_j''}$. The transformation applied to the mean velocity is the one proposed by Ref. [29] for compressible wall-bounded turbulence, denoted as U^{TL} (and hereafter referred to as TL transformation), which has been shown to collapse the mean velocity profile for Mach number up to 4.0 for compressible turbulent channel flows.

The TL transformation naturally yields the semilocal wall normal coordinate y^* defined as [17]

$$y^* = \frac{\overline{\rho} u_\tau^* y}{\overline{\mu}}, \quad u_\tau^* = \sqrt{\tau_w/\overline{\rho}}, \tag{29}$$

where $\overline{\rho}$ and $\overline{\mu}$ are local flow density and dynamic viscosity averaged in time and homogeneous directions; τ_w is the mean wall shear stress. The corresponding normalization of the Reynolds stress is given as $\overline{\rho u_i'' u_j''}/\tau_w$. For reference, the compressible channel flow data by Ref. [31] ($M_b = 1.50$, $\mathrm{Re}_b = 5000$) and the incompressible channel flow data from Ref. [32] are also included.

TABLE II. Grid resolutions in wall units and percentage increase in the wall shear stress for the permeable-wall LES cases. Top and bottom wall statistics are reported separately. The increasing dash spacing in the line style indicates increasing wall permeability. The time step and the CPU hours per flow-through time are also reported.

				Bottom				Top					CPU hours/
Group	M_b	R	Line style	Δx^+	$\Delta y^+_{\min}$	Δz^+	$\Delta\tau_w(\%)$	Δx^+	$\Delta y^+_{\min}$	Δz^+	$\Delta\tau_w(\%)$	Δt	flow-through time
(I)	1.50	∞	———	25.37	0.14	11.30	0.00	25.23	0.14	11.24	0.00	2×10^{-4}	16 457
		1.00	- - - - - -	21.53	0.14	10.77	2.33	20.94	0.14	10.47	+2.18	2×10^{-4}	16 384
		0.50	- - - - .	22.87	0.15	11.44	18.02	21.40	0.14	10.70	+4.42	2×10^{-4}	16 384
(II)	3.50	∞	———	28.85	0.30	16.06	0.00	29.22	0.30	16.26	0.00	2×10^{-4}	82 285
		1.00	- - - - - -	30.22	0.32	17.01	12.35	29.38	0.31	16.53	+3.35	1×10^{-4}	87 711
		0.50	- - - - .	23.81	0.27	13.50	10.93	23.00	0.26	13.04	+0.62	1×10^{-4}	137 143

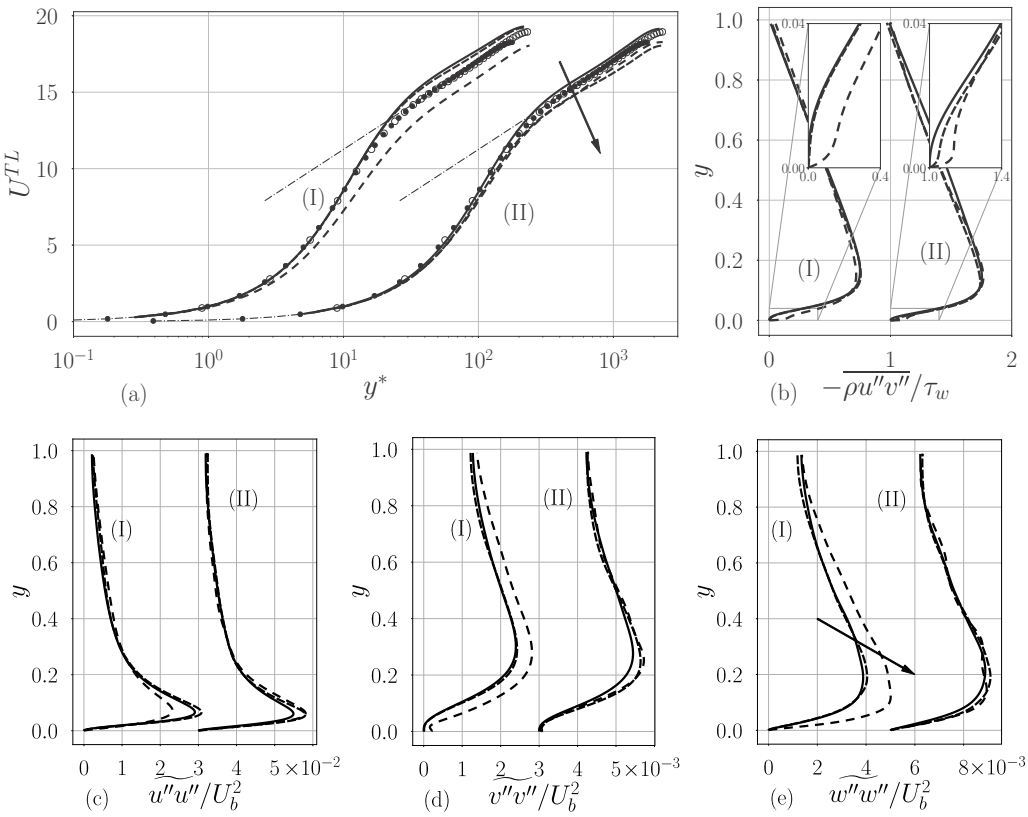

FIG. 2. First and second-order flow statistics: (a) TL-transformed mean streamwise velocity profile [29]. The dashed-dotted line represents the reference log-law $U^{\mathrm{TL}} = 2.5\ln y^* + 5.5$ and the linear law (below $y^* = 5$); (b) Reynolds shear stress normalized by mean wall shear; (c)–(e) Reynolds normal stresses normalized by U_b^2. The line spacing in the plot increases with an increasing permeability (decreasing resistance R).Arrows indicate direction of increasing permeability.

For the LES impermeable wall cases, the near-wall profiles agree well with reference data. The case at $M_b = 1.50$ starts to see an overshoot beyond $y^* \approx 15$, while case with the higher Mach number shows good agreement across the whole channel. With wall permeability, the mean velocity profile starts to deviate from its impermeable reference as shown in Fig. 2(a). For both Mach numbers, the effect of wall permeability is not apparent at $R = 1.00$. As R decreases to the value of 0.50, the mismatch appears early in the viscous sublayer region and continues all the way to the channel center. At such low values of wall acoustic resistance, the typical viscous sublayer scaling, or $U^{TL} \approx y^*$, is no longer followed. For this reason, the value of y^* loses its correlation to a specific state of wall-bounded turbulence. As a result, most of the statistics shown later will simply use the wall normal coordinate y made dimensionless with the channel semi-height δ. The high near-wall velocity gradient indicates the increase in the mean wall shear stress compared to the impermeable wall baselines, whose values are also reported in terms of percentage increase $\Delta\tau_w$ in Table II. Except for the one with $R = 1.00$ at $M_b = 1.50$, all other cases show at least 10% of enhancement in the mean wall shear stress. The Reynolds shear stress $-\overline{\rho u'' v''}/\tau_w$ reflects the near-wall changes more clearly and is given in Fig. 2(b). A local peak of $-\overline{\rho u'' v''}/\tau_w$ is observed below $y = 0.02$ (see the inset view), which means a stronger correlation between fluctuations in streamwise and wall normal velocity components at this location, and thus an enhancement in

the convective momentum transfer in the near-wall region. Again, such enhancement has not been observed in the low permeability case at low Mach number with high resistance.

Figures 2(c) and 2(e) demonstrate the normal Reynolds stresses, which are normalized by U_b^2 to allow for an easier comparison. A redistribution of the turbulence intensities has been observed due to wall porosity, which also varies with Mach numbers. Typically the no-slip impermeable wall conditions lead to a two-component nature of turbulent channel flow in the near-wall region—the vertical component $\widetilde{v''v''}$ grows roughly as y^4 while $\widetilde{u''u''}$ and $\widetilde{w''w''}$ as y^2, resulting in $\widetilde{v''v''}$ approaching zero at the wall much faster than the other two components [33]. In addition, $\widetilde{u''u''}$ is usually the largest component because of the fluctuation energy extraction from the mean streamwise shear. The high resistance cases ($R = 1.00$) at both Mach numbers do not show significant differences with respect to the impermeable baseline cases, except for a slightly increase in the peak of $\widetilde{u''u''}$. For lower resistance $R = 0.50$ at low Mach number $M_b = 1.50$, a decrease is observed in $\widetilde{u''u''}$ in the near-wall region and the other two components exhibit enhancement in the lower half channel. However, when keeping the same resistance and increasing the Mach number, an increase in all stress components are observed. Nonzero values of $\widetilde{v''v''}$ are observed due to the nonzero transpiration velocity v' at the permeable wall. This redistribution of the normal stresses will be addressed with further details in Sec. VI B. The observation in the first and second-order mean statistics suggests that for cases with sufficiently high wall permeability, the near-wall turbulence structures might have been altered. The structural-wise changes are investigated in the next section.

V. NEAR-WALL TURBULENCE STRUCTURES

A. Streamwise traveling structures

The mean flow statistics indicates the possibility of the alterations in near-wall turbulence structures, which is confirmed in Fig. 3, where for cases with $R = 0.50$ the isosurface of Q criterion [34] are shown on top of contour of wall pressure fluctuations.

Groups of streamwise traveling structures are identified, which leave a coherent footprint of pressure waves on the wall surface. The characteristic wavelength of such structures is found to be $\lambda_x \approx 0.47$ for both Mach numbers. The velocity field obtained by subtracting the mean flow from the instantaneous field shows that such waves create localized circulation zones in regions of positive pressure fluctuation p', separated by flow entrainment region that brings the high momentum fluid toward the wall, which leads to a local increase of the wall shear stress. With the same $R = 0.50$, a discernible difference in the pressure pattern of $M_b = 1.50$ and $M_b = 3.50$ is observed. At $M_b = 3.50$, the wave structure is confined closer near the wall, with the high pressure blob more concentrated in a smaller region, as compared to the pattern at $M_b = 1.50$. In Ref. [14], the permeable walls in subsonic channel flows trigger Kelvin-Helmholtz type of instability, and result in a layer of spanwise rollers that disrupt the viscous sublayer. At that low Mach number, the characteristic length scale of the rollers was found to be $\lambda_x \approx 0.40$ with regions of flow entrainment and circulation zones of comparable size. In the current work a longer length scale of the instability is observed with $\lambda_x \approx 0.47$ and circulation regions are more extended than the entrainment region. As the Mach number increases, a deviation from spatially harmonic pattern is observed, with more localized and abrupt positive pressure fluctuations, indicating possible coupling with the overlying flow field, associated to a stronger flow response; For the lower Mach number case, the same value of dimensionless resistance ($R = 0.50$) yields a weaker response, which preserves its modal nature and hence likely decouples from the turbulence. This behavior might relate to the increasing temperature gradient as the Mach number increases in such type of flow, which requires further investigation in the future. For all Mach number cases, the observed waves are 'trapped' as shown in a companion work [16], in the sense of wave being confined near the impedance boundary but able to propagate along the boundary indefinitely. The same type of waves have been defined by Ref. [35] as "surface waves" when discussing waves confined close to an impedance lining in a duct with flow. However,

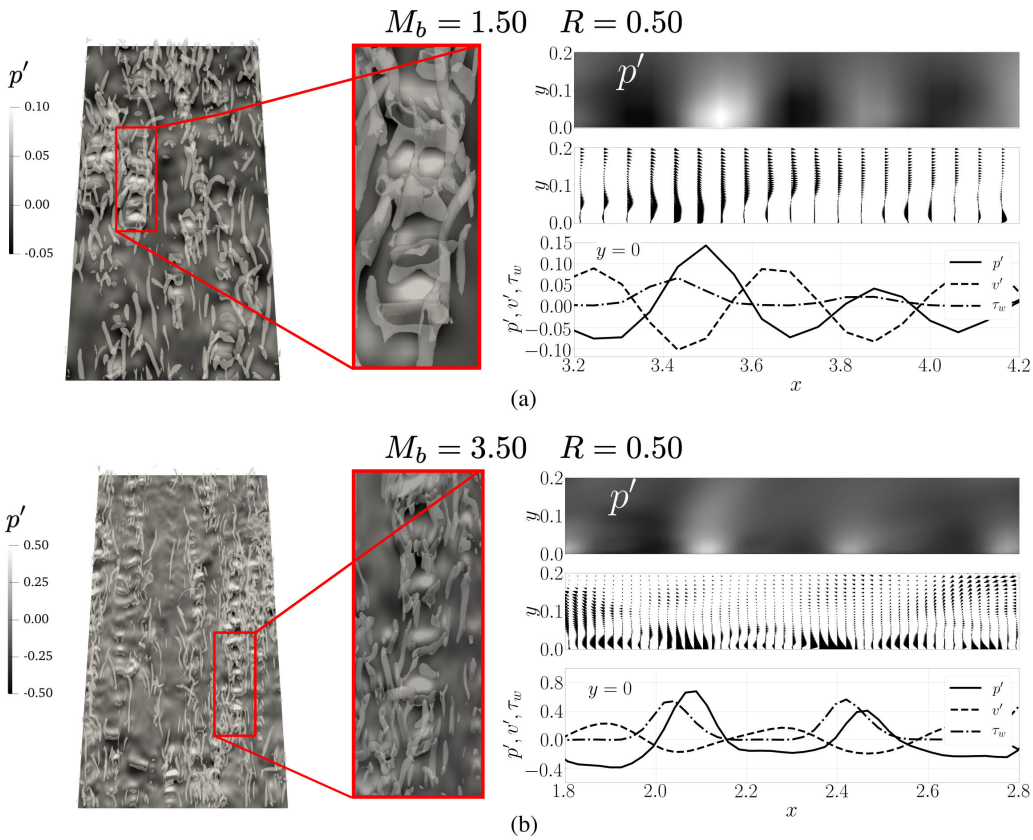

FIG. 3. Near-wall instability waves resulting from the interaction of the impedance wall and the overlying supersonic turbulent flow field. The cases presented are (a) $M_b = 1.50$, $R = 0.50$ and (b) $M_b = 3.50$, $R = 0.50$. In each subfigure, the following quantities are shown: Isosurface of Q criterion overlaid on the pressure fluctuation contour on the wall (left); spanwise slice of the pressure contour showing the characteristic wave length λ_x (top right); The vector plot of fluctuating velocity field (center right); Signals of wall normal velocity, pressure fluctuation and shear stress at $y = 0$ at the same spanwise slice (bottom right).

in acoustics the term "trapped" refers to a wave that is not able to escape a given region in any direction [36], and therefore the word "trapped" is avoided here for clarity.

B. Vorticity fluctuation intensity

The presented flow structures due to the response of permeable walls greatly affect the vorticity field near the wall. The normalized root-mean-square (RMS) vorticity components $\langle\omega_x\rangle^*_{\text{rms}}$ are presented in Fig. 4. The superscript "$*$" indicates the value that is normalized by the quantity $\mu_{\text{ref}}/\tau_{w,R=\infty}$ taken from the baseline cases. The logarithm scale is used to emphasize the near-wall changes.

In all the cases presented, the wall normal component $\langle\omega_y\rangle^*_{\text{rms}}$ does not experience obvious changes over various wall permeabilities except for the low resistance case at $M_b = 1.50$ in which an enhancement of $\langle\omega_y\rangle^*_{\text{rms}}$ is shown below $y = 0.1$. The effect of wall permeability mainly exists in the other two components, $\langle\omega_x\rangle^*_{\text{rms}}$ and $\langle\omega_z\rangle^*_{\text{rms}}$. In a typical impermeable channel flow, clusters of streamwise streaks resides near the wall, which can be approximately visualized as streamwise aligned vortices. On the average, these streaks create a local maximum at $y \approx 0.1$ in $\langle\omega_x\rangle^*_{\text{rms}}$ (roughly at $y^* = 24 \sim 25$ in an impermeable wall case), which corresponds to the averaged height of the